

UCS503: SOFTWARE ENGINEERING

L	T	P	Cr
3	0	2	4.0

Course Objectives: To plan and manage large scale software and learn emerging trends in software engineering.

Software Engineering and Processes: Introduction to Software Engineering, Software Evolution, Software Characteristics, Software Crisis: Problems and Causes, Software process models -Waterfall, Iterative, Incremental and Evolutionary process models.Role of Software in Sustainable Development,Green Coding Guidelines (Optimizing Memory & CPU Utilization), Regulatory Aspects & Global Policies on Green IT. ISO Standards for Green IT

Requirements Engineering: Problem Analysis, Requirement Elicitation and Validation, Requirement Analysis Approaches- Structured Analysis Vs Object Oriented Analysis, Flow modeling through Data Flow Diagram and Data Dictionary, Data Modeling through E-R Diagram, Requirements modeling through UML, based on Scenario, Behavioral and Class modeling, documenting Software Requirement Specification (SRS)

Software Design and construction: System design principles like levels of abstraction, separation of concerns, information hiding, coupling and cohesion, Structured design (top-down or functional decomposition), object-oriented design, event driven design, component-level design, test driven design, data design at various levels, architecture design like Model View Controller, Client – Server architecture. Coding Practices: Techniques, Refactoring, Integration Strategies, Internal Documentation., Green Software Architecture (Microservices vs. Monolith, Edge Computing)

Software Verification and Validation: Levels of Testing, Functional Testing, Structural Testing, Test Plan, Test Case Specification, Software Testing Strategies, Verification & Validation, Unit and Integration Testing, Alpha & Beta Testing, White box and black box testing techniques, System Testing and Overview of Debugging.

Agile Software Development: Agile Manifesto, Twelve Practices of eXtreme Programming (XP), XP values, XP practices, velocity, spikes, working of Scrum, product backlog, sprint backlog, Adaptive Software Development(ASD), Feature Driven Development (FDD), Test Driven Development, Dynamic System Development Method(DSDM), and Crystal Methodology,Agile Requirement and Design: User Stories, Story Boards, UI Sketching and Story Cards. Energy-Aware UI/UX Design (Dark Mode, Reduced Animations, Low Power UX Patterns)

Software Project Management: Overview of Project Management: Scope, Time and Cost estimations. Case Studies: Energy-Efficient Software Practices in Google, Microsoft, Meta

Laboratory work:

Implementation of Software Engineering concepts and exposure to CASE tools like Rational Software Suit through projects.

Course Learning Outcomes (CLOs) / Course Objectives (COs):**On completion of this course, the students will be able to**

1. Analyze software development process models for software development life cycle.
2. Elicit, describe, and evaluate a system's requirements and analyze them using various UML models.
3. Demonstrate the use of design principles in designing data, architecture, user and component level design.
4. Test the system by planning appropriate test cases and applying relevant test strategies.
5. Comprehend the use of agile development methodologies including UI sketching, user stories, story cards and backlog management.

Text Books:

1. Pressman R., Software Engineering, A Practitioner's Approach, McGraw Hill International, 7th ed. (2010).
2. Sommerville I., Software Engineering, Addison-Wesley Publishing Company, 9th ed. (2011).

Reference Books:

1. Jalote P., An integrated Approach to Software Engineering, Narosa, 3rd ed. (2005).
2. Booch G., Rumbaugh J., Jacobson I., The Unified Modeling Language User Guide, 2nd ed. (2005).